DS HW 1

1) Sum of Two Matrices :

```
#include <stdio.h>
int main()
{
    int a[10][10], b[10][10], sum[10][10];
    int r, c, i, j;

    printf("Enter rows and columns: ");
    scanf("%d%d", &r, &c);

    printf("Enter first matrix:\n");
    for(i=0;i<r;i++)
        for(j=0;j<c;j++)
            scanf("%d",&a[i][j]);

    printf("Enter second matrix:\n");
    for(i=0;i<r;i++)
        for(j=0;j<c;j++)
            scanf("%d",&b[i][j]);

    printf("Sum of matrices:\n");
    for(i=0;i<r;i++)
    {
        for(j=0;j<c;j++)
```

```
        {
            sum[i][j]=a[i][j]+b[i][j];
            printf("%d ",sum[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

OUTPUT:

```
Enter rows and columns: 2 2
Enter first matrix:
7 9
4 5
Enter second matrix:
2 6
1 7
Sum of matrices:
9 15
5 12
```

2) Read N Numbers and Display in Reverse Order:

```
#include <stdio.h>
int main()
{
```

```
    int a[100], n, i;

    printf("Enter number of elements: ");
    scanf("%d",&n);

    printf("Enter elements:\n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    printf("Reverse order:\n");
    for(i=n-1;i>=0;i--)
        printf("%d ",a[i]);

    return 0;
}
```

OUTPUT:

```
Enter number of elements: 5
Enter elements:
89 5 10 2 4
Reverse order:
4 2 10 5 89

=== Code Execution Successful ===
```

3) Count Total Number of Duplicate Elements:

```
#include <stdio.h>
int main()
{
   int a[100], n, i, j, count=0;

   printf("Enter size: ");
   scanf("%d",&n);

   printf("Enter elements:\n");
   for(i=0;i<n;i++)
      scanf("%d",&a[i]);

   for(i=0;i<n;i++)
   {
      for(j=i+1;j<n;j++)
      {
         if(a[i]==a[j])
         {
            count++;
            break;
         }
      }
   }

   printf("Duplicate elements = %d",count);
```

```
    return 0;
}
```

OUTPUT:

```
Enter size: 5
Enter elements:
2 5 2 5 2
Duplicate elements = 3
=== Code Execution Successful ===
```

4) Print Unique Elements in an Array:

```
#include <stdio.h>

int main()
{
    int a[100], n, i, j, flag;

    printf("Enter size: ");
    scanf("%d",&n);

    printf("Enter elements:\n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    printf("Unique elements:\n");

    for(i=0;i<n;i++)
```

```
    {
        flag=1;

        for(j=0;j<n;j++)
        {
            if(i!=j && a[i]==a[j])
            {
                flag=0;
                break;
            }
        }

        if(flag)
            printf("%d ",a[i]);
    }

    return 0;
}
```

OUTPUT:

```
Enter size: 7
Enter elements:
5 2 8 2 0 8 7
Unique elements:
5 0 7
=== Code Execution Successful ===
```

5) Insert an Element in a Sorted Array:

```
#include <stdio.h>

int main()
{
   int a[100], n, item, i;

   printf("Enter size: ");
   scanf("%d",&n);

   printf("Enter sorted elements:\n");
   for(i=0;i<n;i++)
      scanf("%d",&a[i]);

   printf("Enter element to insert: ");
   scanf("%d",&item);

   i=n-1;

   while(i>=0 && a[i]>item)
   {
      a[i+1]=a[i];
      i--;
   }

   a[i+1]=item;
   n++;
```

```
    printf("Array after insertion:\n");

    for(i=0;i<n;i++)
        printf("%d ",a[i]);

    return 0;
}
```

OUTPUT:

```
Enter size: 5
Enter sorted elements:
23 25 26 27 28
Enter element to insert: 24
Array after insertion:
23 24 25 26 27 28
=== Code Execution Successful ===
```

6) Insert an Element in an Unsorted Array:

```
#include <stdio.h>

int main()
{
    int a[100], n, item, pos, i;

    printf("Enter size: ");
    scanf("%d",&n);
```

```
    printf("Enter elements:\n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    printf("Enter position: ");
    scanf("%d",&pos);

    printf("Enter element: ");
    scanf("%d",&item);

    for(i=n;i>=pos;i--)
        a[i]=a[i-1];

    a[pos-1]=item;
    n++;

    printf("Array after insertion:\n");

    for(i=0;i<n;i++)
        printf("%d ",a[i]);

    return 0;
}
```

OUTPUT:

Enter size: 6

Enter elements:

10 30 50 70 90 100

Enter position: 3

Enter element: 40

Array after insertion:

10 30 40 50 70 90 100

=== Code Execution Successful ===

7) Matrix Multiplication:

```
#include <stdio.h>

int main()
{
   int a[10][10], b[10][10], c[10][10];
   int r1,c1,r2,c2,i,j,k;

   printf("Enter rows and columns of first matrix: ");
   scanf("%d%d",&r1,&c1);

   printf("Enter rows and columns of second matrix: ");
   scanf("%d%d",&r2,&c2);

   if(c1!=r2)
   {
      printf("Multiplication not possible");
      return 0;
   }

   printf("Enter first matrix:\n");
```

```
for(i=0;i<r1;i++)
    for(j=0;j<c1;j++)
        scanf("%d",&a[i][j]);

printf("Enter second matrix:\n");
for(i=0;i<r2;i++)
    for(j=0;j<c2;j++)
        scanf("%d",&b[i][j]);

for(i=0;i<r1;i++)
{
    for(j=0;j<c2;j++)
    {
        c[i][j]=0;
        for(k=0;k<c1;k++)
            c[i][j]+=a[i][k]*b[k][j];
    }
}

printf("Result:\n");

for(i=0;i<r1;i++)
{
    for(j=0;j<c2;j++)
        printf("%d ",c[i][j]);
    printf("\n");
}
```

```
    return 0;
}
```

OUTPUT:

```
Enter rows and columns of first matrix: 2 2
Enter rows and columns of second matrix: 2 2
Enter first matrix:
1 2
3 4
Enter second matrix:
5 6
7 8
Result:
19 22
43 50
=== Code Execution Successful ===
```

8) Matrix Transpose:

```
#include <stdio.h>

int main()
{
    int a[10][10];
    int r,c,i,j;

    printf("Enter rows and columns: ");
    scanf("%d%d",&r,&c);
```

```
    printf("Enter matrix:\n");

    for(i=0;i<r;i++)
        for(j=0;j<c;j++)
            scanf("%d",&a[i][j]);

    printf("Transpose:\n");

    for(i=0;i<c;i++)
    {
        for(j=0;j<r;j++)
            printf("%d ",a[j][i]);
        printf("\n");
    }

    return 0;
}
```

OUTPUT:

```
Enter rows and columns: 2 3
Enter matrix:
1 2 3
4 5 6
Transpose:
1 4
2 5
3 6
```

9) Right Diagonal Sum:

```
#include <stdio.h>

int main()
{
   int a[10][10], n, i, sum=0;

   printf("Enter order of matrix: ");
   scanf("%d",&n);

   printf("Enter matrix:\n");

   for(i=0;i<n;i++)
      for(int j=0;j<n;j++)
         scanf("%d",&a[i][j]);

   for(i=0;i<n;i++)
      sum+=a[i][n-1-i];

   printf("Right diagonal sum = %d",sum);

   return 0;
}
```

OUTPUT:

```
Enter order of matrix: 3
Enter matrix:
1 2 3
```

```
4 5 6
7 8 9
Right diagonal sum = 15
=== Code Execution Successful ===
```

10) Check Whether Matrix is Identity Matrix:

```
#include <stdio.h>

int main()
{
    int a[10][10], n, i, j, flag=1;

    printf("Enter order of matrix: ");
    scanf("%d",&n);

    printf("Enter matrix:\n");

    for(i=0;i<n;i++)
        for(j=0;j<n;j++)
            scanf("%d",&a[i][j]);

    for(i=0;i<n;i++)
    {
        for(j=0;j<n;j++)
        {
            if(i==j && a[i][j]!=1)
                flag=0;
```

```
            if(i!=j && a[i][j]!=0)
                flag=0;
        }
    }

    if(flag)
        printf("Identity Matrix");
    else
        printf("Not an Identity Matrix");

    return 0;
}
```

OUTPUT:

```
Enter order of matrix: 3
Enter matrix:
1 0 0
0 1 0
0 0 1
Identity Matrix
=== Code Execution Successful ===
```

11) Count Vowels, Consonants, Digits and White Spaces:

```
#include <stdio.h>

int main()
{
    char str[100];
    int i,v=0,c=0,d=0,s=0;
```

```
    printf("Enter a string: ");
    fgets(str,100,stdin);

    for(i=0;str[i]!='\0';i++)
    {
        if(str[i]=='a'||str[i]=='e'||str[i]=='i'||str[i]=='o'||str[i]=='u'||
           str[i]=='A'||str[i]=='E'||str[i]=='I'||str[i]=='O'||str[i]=='U')
            v++;
        else if((str[i]>='A'&&str[i]<='Z')||(str[i]>='a'&&str[i]<='z'))
            c++;
        else if(str[i]>='0'&&str[i]<='9')
            d++;
        else if(str[i]==' ')
            s++;
    }

    printf("Vowels=%d\nConsonants=%d\nDigits=%d\nWhite spaces=%d",v,c,d,s);

    return 0;
}
```

OUTPUT:

Enter a string: Hello 123

Vowels=2

Consonants=3

Digits=3

White spaces=1

=== Code Execution Successful ===

12) Frequency of a Character in a String:

```
#include <stdio.h>

int main()
{
   char str[100], ch;
   int i,count=0;

   printf("Enter string: ");
   fgets(str,100,stdin);

   printf("Enter character: ");
   scanf("%c",&ch);

   for(i=0;str[i]!='\0';i++)
      if(str[i]==ch)
         count++;

   printf("Frequency = %d",count);

   return 0;
}
```

OUTPUT:

Enter string: banana

Enter character: a

Frequency = 3

=== Code Execution Successful ===

13) Count Words in a String:

```
#include <stdio.h>

int main()
{
   char str[100];
   int i,count=1;

   printf("Enter string: ");
   fgets(str,100,stdin);

   for(i=0;str[i]!='\0';i++)
   {
      if(str[i]==' ')
         count++;
   }

   printf("Words = %d",count);

   return 0;
}
```

OUTPUT:

Enter string: Anna University Chennai

Words = 3

=== Code Execution Successful ===

14) Sort String Array:

```
#include <stdio.h>
#include <string.h>

int main()
{
    char str[10][20], temp[20];
    int n,i,j;

    printf("Enter number of strings: ");
    scanf("%d",&n);

    printf("Enter strings:\n");

    for(i=0;i<n;i++)
        scanf("%s",str[i]);

    for(i=0;i<n-1;i++)
    {
        for(j=i+1;j<n;j++)
        {
            if(strcmp(str[i],str[j])>0)
            {
                strcpy(temp,str[i]);
                strcpy(str[i],str[j]);
```

```
            strcpy(str[j],temp);
        }
    }
}

printf("Sorted strings:\n");

for(i=0;i<n;i++)
    printf("%s\n",str[i]);

return 0;
}
```

OUTPUT:

```
Enter number of strings: 5
Enter strings:
orange
apple
grapes
banana
mango
Sorted strings:
apple
banana
grapes
mango
orange
 === Code Execution Successful ===
```

15) Toggle Case of a Sentence:

```
#include <stdio.h>

int main()
{
   char str[100];
   int i;

   printf("Enter a sentence: ");
   fgets(str,100,stdin);

   for(i=0;str[i]!='\0';i++)
   {
      if(str[i]>='A'&&str[i]<='Z')
         str[i]=str[i]+32;
      else if(str[i]>='a'&&str[i]<='z')
         str[i]=str[i]-32;
   }

   printf("After toggle:\n%s",str);

   return 0;
}
```

OUTPUT:

Enter a sentence: Hello World

After toggle:

hELLO wORLD

=== Code Execution Successful ===